Multiple-Input, Multiple-Output Systems (MIMO)

State-space Models, Linearization, Dynamic Simulation in Simulink

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Outline

How to formulate MIMO models?

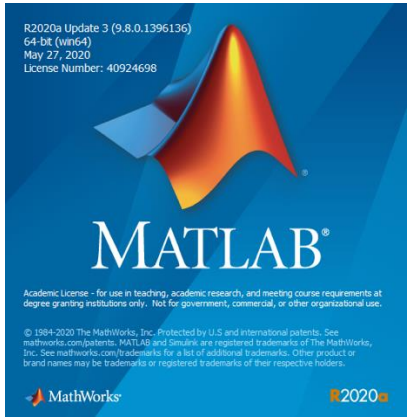
- General Form of a DAE Model
- Model Linearization
- State-space (SS) Models

How to simulate MIMO models?

- Time-Domain Solution of SS Models
- MIMO Systems in MATLAB Simulink

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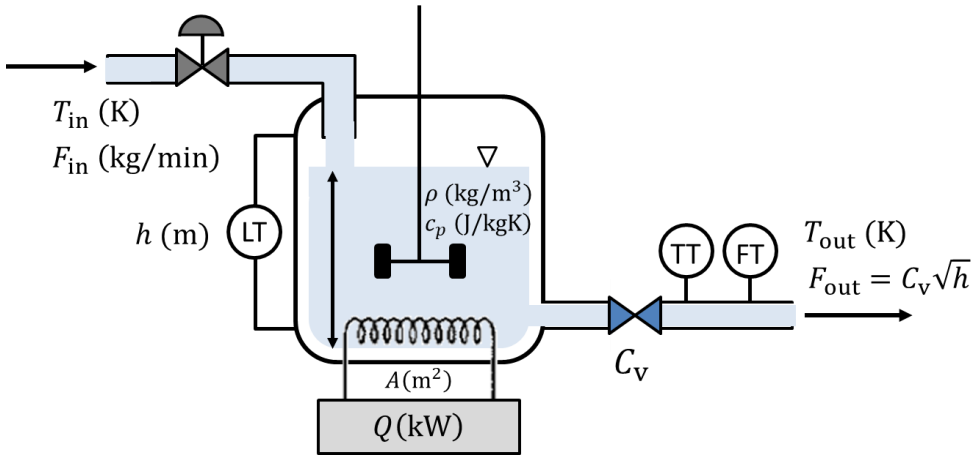
MATLAB Simulink



First, let's learn **MATLAB Simulink** for Process Dynamic Simulation.

We'll use the Heated Stirred Tank example for:

- Dynamic Simulation
- Model Linearization
- Generating State-space Models



$$\frac{dh}{dt} = \frac{1}{\rho A}(F_{in} - F_{out})$$

$$\frac{dT_{out}}{dt} = \frac{F_{in}}{\rho A h}(T_{in} - T_{out}) + \frac{Q}{\rho c_p A h}$$

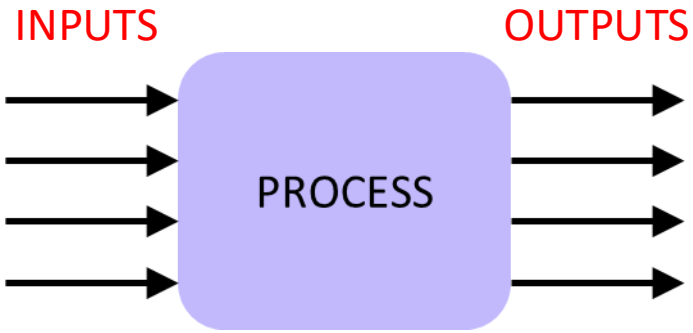
$$F_{out} = C_v\sqrt{h}$$

Variable	Description	Nominal Value
Inputs		
F_{in}	Flow rate of inlet	200 kg/min
T_{in}	Temperature of inlet	298 K
Q	Rate of heat supplied	100 kW
Outputs		
F_{out}	Flow rate of outlet	200 kg/min
T_{out}	Temperature of outlet	305.17 K
h	Liquid height in the tank	2.00 m
Constants		
ρ	Density	1000 kg/m^3
c_p	Specific heat capacity	4184 J/kgK
A	Base area of tank	1.00 m^2
C_v	Valve coefficient	141.42 kg/min/m$^{0.5}$

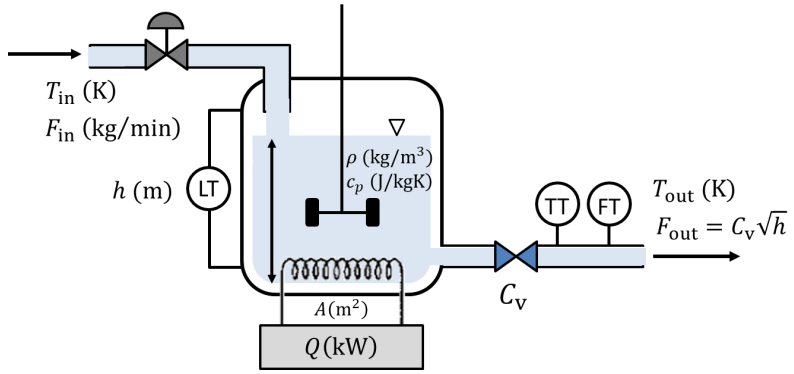
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MIMO Processes

- **Recall:** Real-world processes are naturally MIMO Processes.
- Each input can affect each output, thus all variables may be *interacting* with each other.
- Today, we'll discuss how to represent a MIMO process as a *state-space model*.



Example: Heated Stirred Tank with No Control



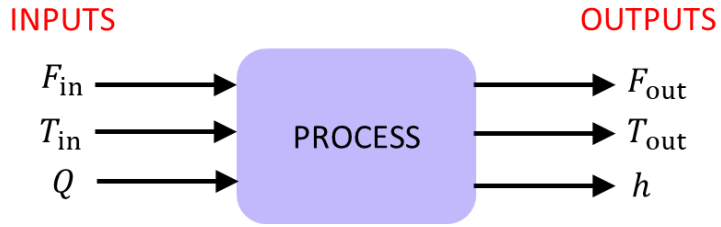
Assumptions:

1. Perfect mixing
2. Properties ρ and c_p are the same in all streams.
3. Negligible heat losses.

$$\frac{dh}{dt} = \frac{1}{\rho A}(F_{in} - F_{out})$$

$$\frac{dT_{out}}{dt} = \frac{F_{in}}{\rho A h}(T_{in} - T_{out}) + \frac{Q}{\rho c_p A h}$$

$$F_{out} = C_v\sqrt{h}$$



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General Form of DAE

General form of the DAE model:

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t))$$
$$\mathbf{y} = \mathbf{g}(t, \mathbf{x}(t), \mathbf{u}(t))$$

If we expand all vectors:

$$\dot{x}_1 = \frac{dx_1}{dt} = f_1(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$

$$\dot{x}_2 = \frac{dx_2}{dt} = f_2(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$

$$\vdots$$

$$y_1 = g_1(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$

$$y_2 = g_2(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$

$$\vdots$$

Models with higher-order derivatives $\left(\frac{d^n x}{dt^n}\right)$ can be cast into a **system of purely first-order DEs** using a change of variables (recall ES 204).

Where:

$\mathbf{x}(t) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix}$ = state variables

(those that appear in *time-derivative* terms)

$\mathbf{u}(t) = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \end{bmatrix}$ = input variables

(those that can be independently *adjusted*)

$\mathbf{y}(t) = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \end{bmatrix}$ = output variables

(those that are physically *measured*)

$\mathbf{f}(\cdot)$ = state equations

(expressions found in the differential equations)

$\mathbf{g}(\cdot)$ = output equations

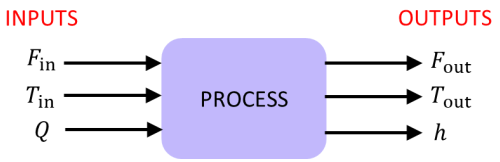
(expressions for computing the output variables)

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General Form of DAE

Example: Heated Stirred Tank with No Control

$$\frac{dx_1}{dt}\ \frac{dh}{dt} = \underbrace{\frac{1}{\rho A}(F_{\text{in}} - F_{\text{out}})}_{f_1(t,\boldsymbol{x},\boldsymbol{u})}$$



$$\frac{dx_2}{dt}\ \frac{dT_{\text{out}}}{dt} = \underbrace{\frac{F_{\text{in}}}{\rho A h}(T_{\text{in}} - T_{\text{out}}) + \frac{Q}{\rho c_p A h}}_{f_2(t,\boldsymbol{x},\boldsymbol{u})}$$

$$y_1\ \ F_{\text{out}} = \underbrace{C_{\text{v}}\sqrt{h}}_{g_1(t,\boldsymbol{x},\boldsymbol{u})}$$

State variables: $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} h \\ T_{\text{out}} \end{bmatrix}$

Input variables: $\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} F_{\text{in}} \\ T_{\text{in}} \\ Q \end{bmatrix}$

Output variables: $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} F_{\text{out}} \\ T_{\text{out}} \\ h \end{bmatrix}$

*Variables can be written in any order inside a vector, but you must consistently follow the same order all the time.

General form of the DAE model:

$$\dot{x}_1 = \frac{dx_1}{dt} = f_1(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$
$$\dot{x}_2 = \frac{dx_2}{dt} = f_2(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$
$$\vdots$$
$$y_1 = g_1(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$
$$y_2 = g_2(t, x_1(t), x_2(t), \dots, u_1(t), u_2(t), \dots)$$
$$\vdots$$

Two Main Kinds of DAE Models:

Linear Model

- All $\boldsymbol{f}$ and $\boldsymbol{g}$ are **linear functions** of $\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{y}$.
- Obeys the superposition principle.
- Its general solution is known and easily computable.

Nonlinear Model

- At least one among $\boldsymbol{f}$, $\boldsymbol{g}$ is a **nonlinear function** of $\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{y}$.
- Its general solution is not available.
- Common nonlinear terms:
  - sqrt
  - products of variables
  - exponents greater than 1
  - sin, cos, tan
  - division of variables
  - log, exp, min, max

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The State-space (or phase-space)

- The state-space is the space of all possible states.
- It allows for the geometric analysis of process dynamics.

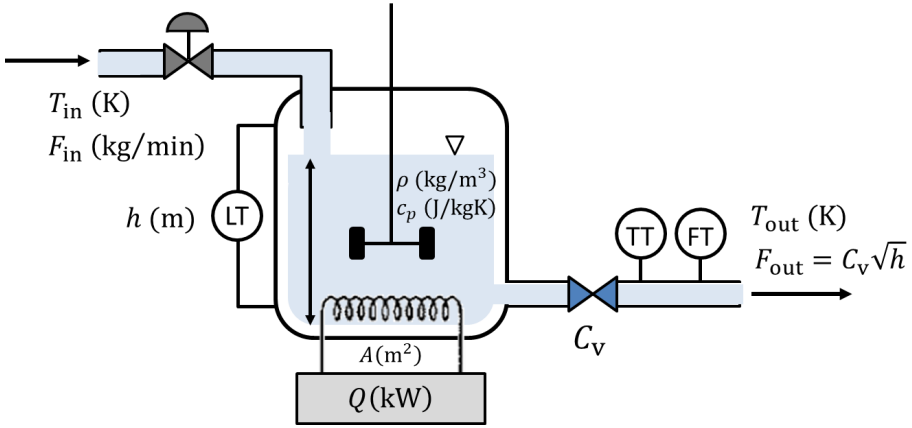
State variables: $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} h \\ T_{\text{out}} \end{bmatrix}$

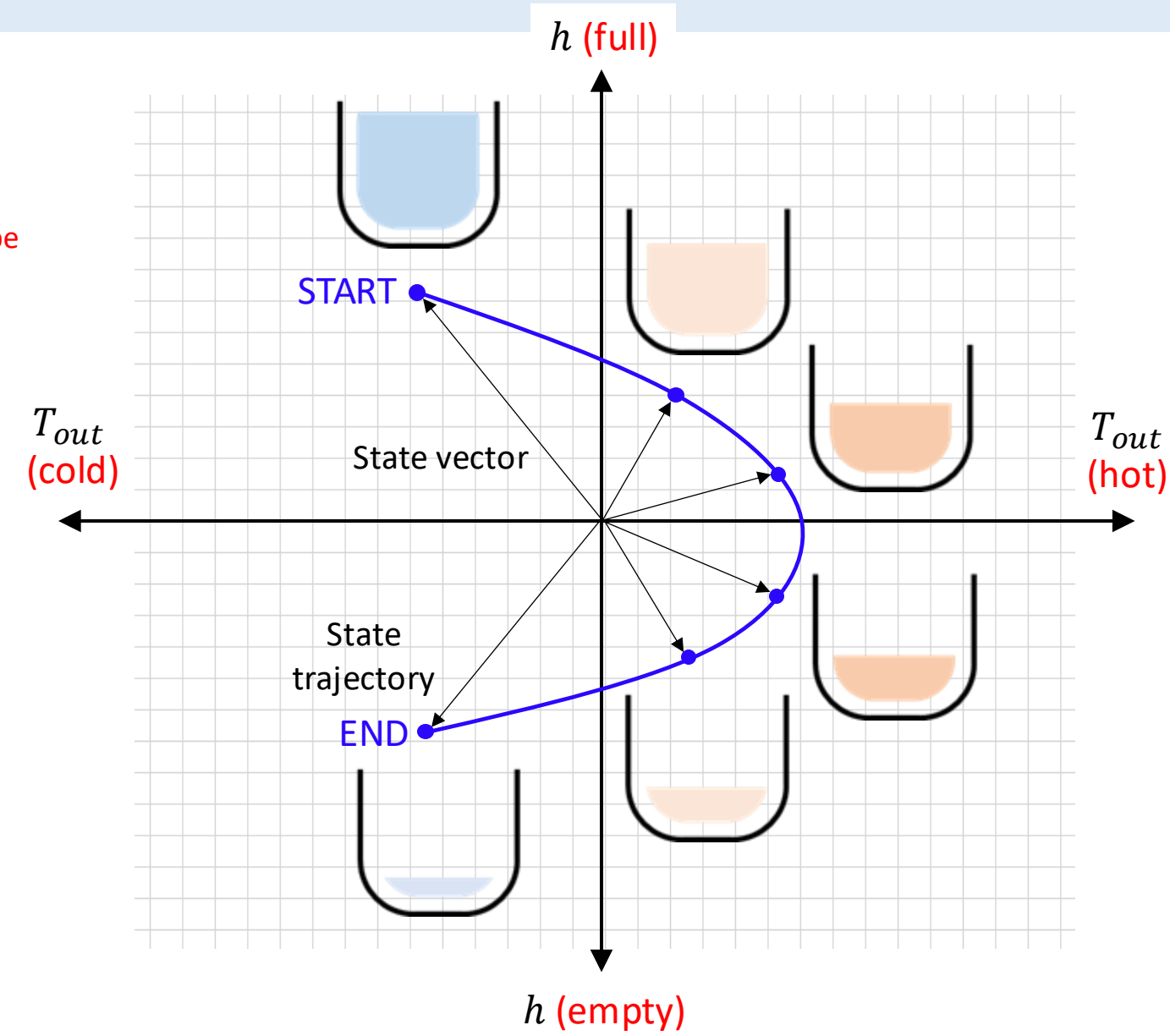
May not be observable, but sufficient to describe the state of the system.

Input variables: $\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} F_{\text{in}} \\ T_{\text{in}} \\ Q \end{bmatrix}$

Output variables: $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} F_{\text{out}} \\ T_{\text{out}} \\ h \end{bmatrix}$

Observable





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Model Linearization

Given a nonlinear DAE, we can **linearize** it using *Taylor series approximation* around an operating point (trimming point).

$$\frac{dh}{dt} = \frac{1}{\rho A}(F_{\text{in}} - F_{\text{out}})$$

$$\frac{dT_{\text{out}}}{dt} = \frac{F_{\text{in}}}{\rho A h}(T_{\text{in}} - T_{\text{out}}) + \frac{Q}{\rho c_p A h}$$

$$F_{\text{out}} = C_{\text{v}}\sqrt{h}$$

NONLINEAR

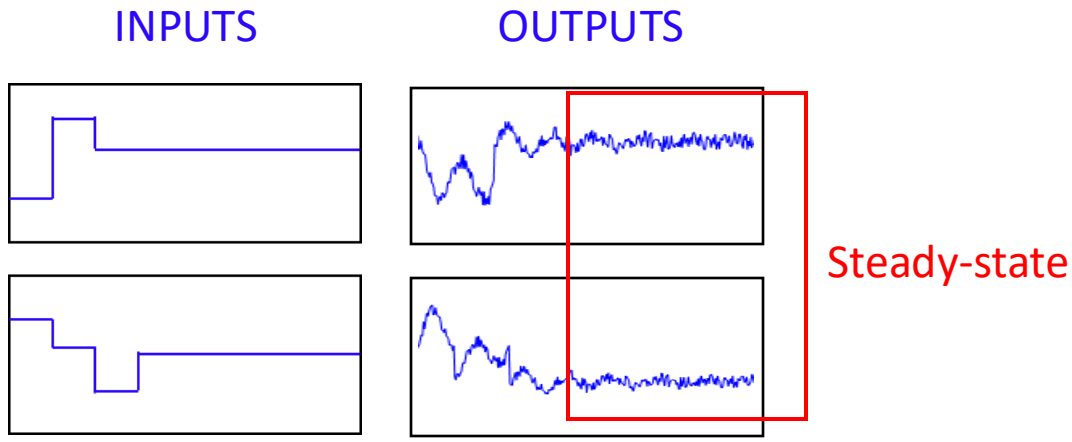
Model Linearization (Taylor series approximation)

$$\frac{dh}{dt} = a_1\, F_{\text{in}} + a_2\, F_{\text{out}}$$

$$\frac{dT_{\text{out}}}{dt} = b_1\, F_{\text{in}} + b_2\, T_{\text{in}} + b_3\, T_{\text{out}} + b_4\, Q + b_5\, h$$

$$F_{\text{out}} = c_1\, h$$

LINEAR



Model linearization depends on the chosen steady-state.

Steady-state

- a condition of the system where all variables are "*virtually constant*" through time.
- In this condition, the values of all variables satisfy the steady-state DAE model (all time-derivatives are zero).

Operating Point

- Any point in the state-space that satisfies the steady-state DAE model.

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Model Linearization

Given a nonlinear DAE, we can **linearize** it using *Taylor series approximation* around an operating point (trimming point).

$$\frac{dh}{dt} = \frac{1}{\rho A}(F_{\text{in}} - F_{\text{out}})$$

$$\frac{dT_{\text{out}}}{dt} = \frac{F_{\text{in}}}{\rho A h}(T_{\text{in}} - T_{\text{out}}) + \frac{Q}{\rho c_p A h}$$

$$F_{\text{out}} = C_v\sqrt{h}$$

NONLINEAR

Model Linearization (Taylor series approximation)

$$\frac{dh}{dt} = a_1 F_{\text{in}} + a_2 F_{\text{out}}$$

$$\frac{dT_{\text{out}}}{dt} = b_1 F_{\text{in}} + b_2 T_{\text{in}} + b_3 T_{\text{out}} + b_4 Q + b_5 h$$

$$F_{\text{out}} = c_1 h$$

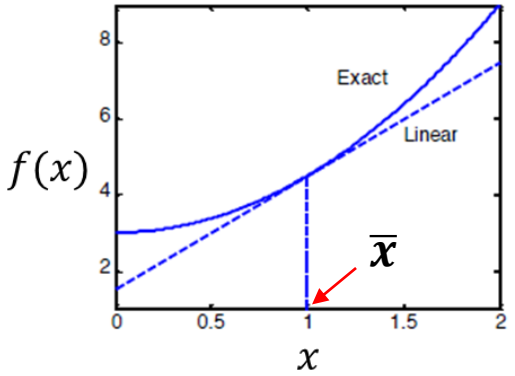
LINEAR

Review of Taylor Series Approximation

A nonlinear function $f(x)$ can be approximated by its first order (linear) Taylor series expansion about a point $\bar{x}$, if the first order derivative exists at $\bar{x}$.

For functions of a **single** variable, x:

$$f(x) \approx f(\bar{x}) + \left.\frac{df}{dx}\right|_{x=\bar{x}} (x - \bar{x})$$



For functions of **two** variables, x_1, x_2 :

$$f(x_1, x_2) \approx f(\bar{x}_1, \bar{x}_2) + \left.\frac{\partial f}{\partial x_1}\right|_{\bar{x}_1,\bar{x}_2} (x_1 - \bar{x}_1) + \left.\frac{\partial f}{\partial x_2}\right|_{\bar{x}_1,\bar{x}_2} (x_2 - \bar{x}_2)$$

For convenience, let's define *deviation variables* as:

$$x' = x - \bar{x}$$

$$f' = f(x) - f(\bar{x})$$

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Model Linearization

Given a nonlinear DAE, we can **linearize** it using *Taylor series approximation* around an operating point (trimming point).

Step 1. Choose / find an operating point.

$$\frac{dh}{dt} = \frac{1}{\rho A}(F_{\text{in}} - F_{\text{out}})$$

$$\frac{dT_{\text{out}}}{dt} = \frac{F_{\text{in}}}{\rho A h}(T_{\text{in}} - T_{\text{out}}) + \frac{Q}{\rho c_p A h}$$

$$F_{\text{out}} = C_{\text{v}}\sqrt{h}$$

Variable	Description	Nominal Value
Inputs		
F_{in}	Flow rate of inlet	200 kg/min
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Outputs		
F_{out}	Flow rate of outlet	200 kg/min
T_{out}	Temperature of outlet	305.17 K
h	Liquid height in the tank	2.00 m
Constants		
ρ	Density	1000 kg/m^3
c_p	Specific heat capacity	4184 J/kgK
A	Base area of tank	1.00 m^2
C_{v}	Valve coefficient	141.42 kg/min/m$^{0.5}$

Step 2. Approximate nonlinear terms using Taylor series

$$\frac{dh'}{dt} = \frac{1}{\rho A}F'_{\text{in}} - \frac{1}{\rho A}C_{\text{v}}\left(\frac{1}{2\sqrt{\bar{h}}}\right)h' = 0.001\,F'_{\text{in}} - 0.05\,h'$$

$$\frac{dT'_{\text{out}}}{dt} = \left[\frac{1}{\rho A\bar{h}}(\bar{T}_{\text{in}} - \bar{T}_{\text{out}})\right]F'_{\text{in}} + \left(\frac{\bar{F}_{\text{in}}}{\rho A\bar{h}}\right)T'_{\text{in}} - \left(\frac{\bar{F}_{\text{in}}}{\rho A\bar{h}}\right)T'_{\text{out}}$$

$$+\left(\frac{1}{\rho c_p A\bar{h}}\right)Q' - \left[\frac{\bar{F}_{\text{in}}(\bar{T}_{\text{in}} - \bar{T}_{\text{out}})}{\rho A\bar{h}^2}\right]h' - \left(\frac{\bar{Q}}{\rho c_p A\bar{h}^2}\right)h'$$

$$\frac{dT'_{\text{out}}}{dt} = -0.0036\,F'_{\text{in}} + 0.1\,T'_{\text{in}} - 0.1\,T'_{\text{out}} + 0.0072\,Q'$$

$$F'_{\text{out}} = C_{\text{v}}\left(\frac{1}{2\sqrt{\bar{h}}}\right)h' = 50\,h'$$

Note: All variables with an overbar (e.g. $\bar{h}$) means that they are fixed at their nominal value (operating point).

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State-space Models

At this point, the linearized model at the given operating point is as follows:

$$\frac{dh'}{dt} = 0.001\,F'_{\text{in}} - 0.05\,h'$$

$$\frac{dT'_{\text{out}}}{dt} = -0.0036\,F'_{\text{in}} + 0.1\,T'_{\text{in}} - 0.1\,T'_{\text{out}} + 0.0072\,Q'$$

$$F'_{\text{out}} = 50\,h'$$

Sizes:

State variables: $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} h \\ T_{\text{out}} \end{bmatrix}$ $n = 2$

Input variables: $\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} F_{\text{in}} \\ T_{\text{in}} \\ Q \end{bmatrix}$ $m = 3$

Output variables: $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} F_{\text{out}} \\ T_{\text{out}} \\ h \end{bmatrix}$ $p = 3$

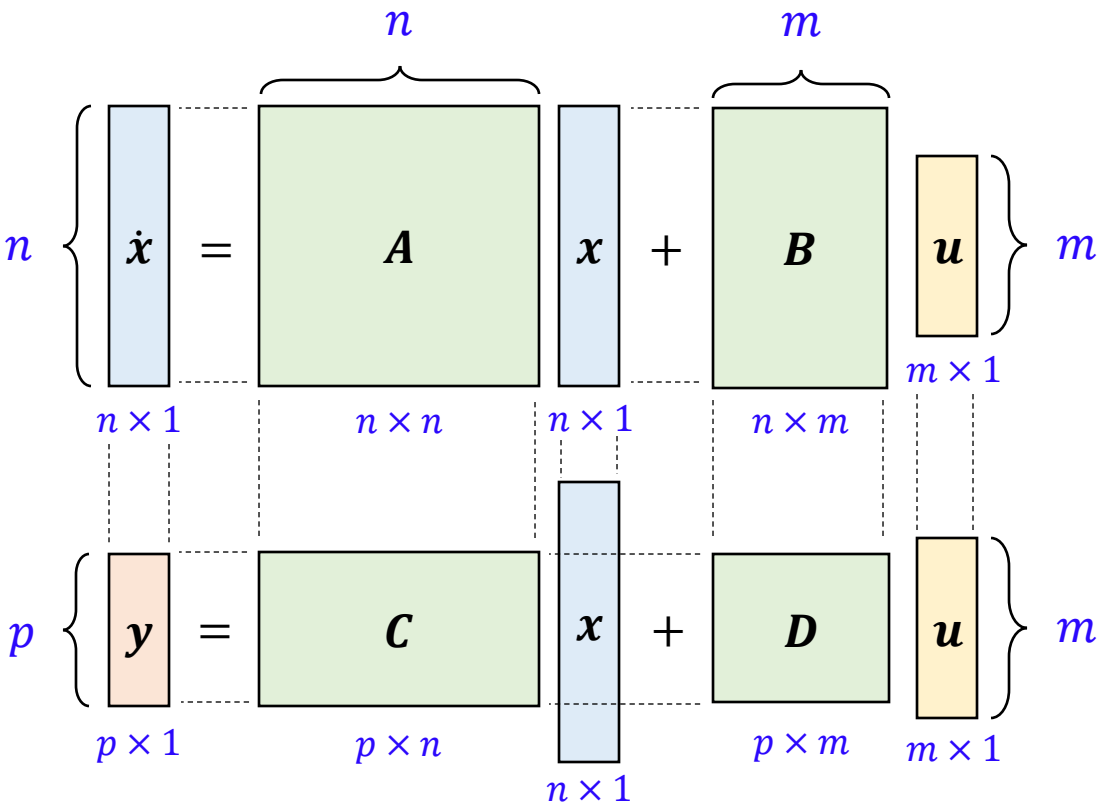
Linear Time-Invariant (LTI) State-space Model

A linear time-invariant state-space model can be written as:

(state equation) $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$

(output equation) $\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u}$

Any system of linear ODEs can be written this way!!



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State-space Models

At this point, the linearized model at the given operating point is as follows:

$$\frac{dh'}{dt} = 0.001\,F'_{\text{in}} - 0.05\,h'$$

$$\frac{dT'_{\text{out}}}{dt} = -0.0036\,F'_{\text{in}} + 0.1\,T'_{\text{in}} - 0.1\,T'_{\text{out}} + 0.0072\,Q'$$

$$F'_{\text{out}} = 50\,h'$$

Sizes:

State variables: $\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} h \\ T_{\text{out}} \end{bmatrix}$ $n = 2$

Input variables: $\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} F_{\text{in}} \\ T_{\text{in}} \\ Q \end{bmatrix}$ $m = 3$

Output variables: $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} F_{\text{out}} \\ T_{\text{out}} \\ h \end{bmatrix}$ $p = 3$

Linear Time-Invariant (LTI) State-space Model

A linear time-invariant state-space model can be written as:

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$$
$$\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u}$$

Example:

$$\begin{bmatrix} \frac{dh'}{dt} \\ \frac{dT'_{\text{out}}}{dt} \end{bmatrix} = \begin{bmatrix} -0.05 & 0 \\ 0 & -0.1 \end{bmatrix} \begin{bmatrix} h' \\ T'_{\text{out}} \end{bmatrix} + \begin{bmatrix} 0.001 & 0 & 0 \\ -0.0036 & 0.1 & 0.0072 \end{bmatrix} \begin{bmatrix} F'_{\text{in}} \\ T'_{\text{in}} \\ Q' \end{bmatrix}$$

$$\begin{bmatrix} F'_{\text{out}} \\ T'_{\text{out}} \\ h' \end{bmatrix} = \begin{bmatrix} 50 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} h' \\ T'_{\text{out}} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} F'_{\text{in}} \\ T'_{\text{in}} \\ Q' \end{bmatrix}$$

State-space Matrices: ***(A, B, C, D)***

$$\boldsymbol{A} = \begin{bmatrix} -0.05 & 0 \\ 0 & -0.1 \end{bmatrix} \quad \boldsymbol{B} = \begin{bmatrix} 0.001 & 0 & 0 \\ -0.0036 & 0.1 & 0.0072 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} 50 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \boldsymbol{D} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

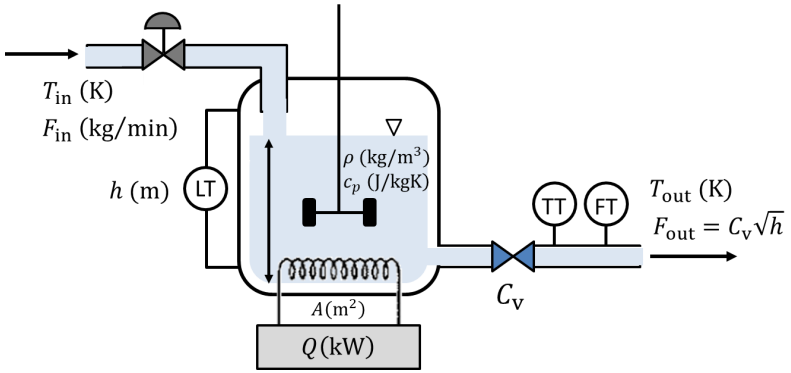
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State-space Models in MATLAB

This is how you can enter a state-space model in MATLAB:

$$\begin{bmatrix} \frac{dh'}{dt} \\ \frac{dT'_{out}}{dt} \end{bmatrix} = \begin{bmatrix} -0.05 & 0 \\ 0 & -0.1 \end{bmatrix} \begin{bmatrix} h' \\ T'_{out} \end{bmatrix} + \begin{bmatrix} 0.001 & 0 & 0 \\ -0.0036 & 0.1 & 0.0072 \end{bmatrix} \begin{bmatrix} F'_{in} \\ T'_{in} \\ Q' \end{bmatrix}$$

$$\begin{bmatrix} F'_{out} \\ T'_{out} \\ h' \end{bmatrix} = \begin{bmatrix} 50 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} h' \\ T'_{out} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} F'_{in} \\ T'_{in} \\ Q' \end{bmatrix}$$




```
>> A = [-0.05 0; 0 -0.1];
>> B = [0.001 0 0; -0.003585 0.1 0.0072];
>> C = [50 0; 0 1; 1 0];
>> D = zeros(3);
>> sys = ss(A,B,C,D)

sys =

  A =
         x1     x2
   x1  -0.05      0
   x2      0   -0.1

  B =
              u1          u2          u3
   x1      0.001           0           0
   x2  -0.003585         0.1      0.0072

  C =
       x1  x2
   y1  50   0
   y2   0   1
   y3   1   0

  D =
       u1  u2  u3
   y1   0   0   0
   y2   0   0   0
   y3   0   0   0

Continuous-time state-space model.

fx >>
```

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MATLAB `step` function

Exercise:

Using the linear state-space model of the Heated Stirred Tank, simulate the response of F_{out}, T_{out}, and h to a step change of +10 kg/min in F_{in}.

$$\begin{bmatrix} \frac{dh'}{dt} \\ \frac{dT'_{out}}{dt} \end{bmatrix} = \begin{bmatrix} -0.05 & 0 \\ 0 & -0.1 \end{bmatrix} \begin{bmatrix} h' \\ T'_{out} \end{bmatrix} + \begin{bmatrix} 0.001 & 0 & 0 \\ -0.0036 & 0.1 & 0.0072 \end{bmatrix} \begin{bmatrix} F'_{in} \\ T'_{in} \\ Q' \end{bmatrix}$$

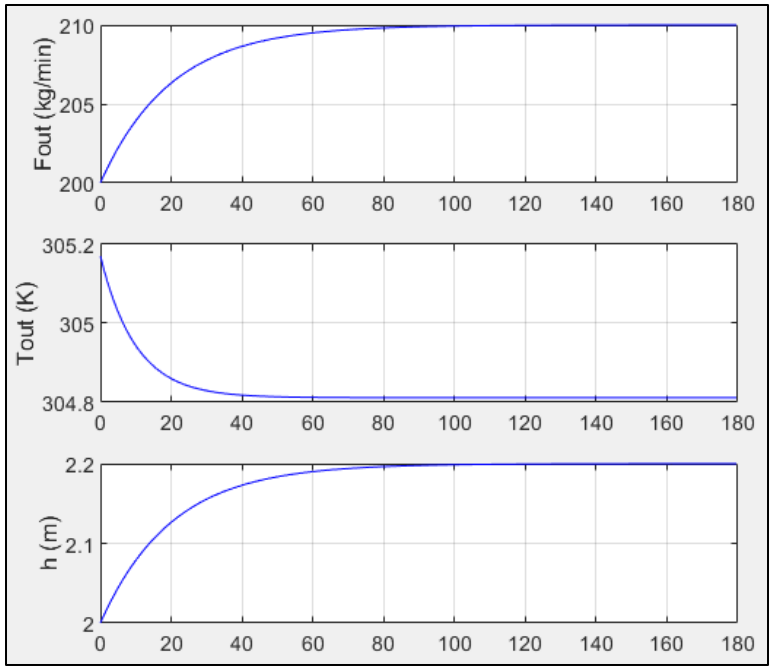
$$\begin{bmatrix} F'_{out} \\ T'_{out} \\ h' \end{bmatrix} = \begin{bmatrix} 50 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} h' \\ T'_{out} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} F'_{in} \\ T'_{in} \\ Q' \end{bmatrix}$$

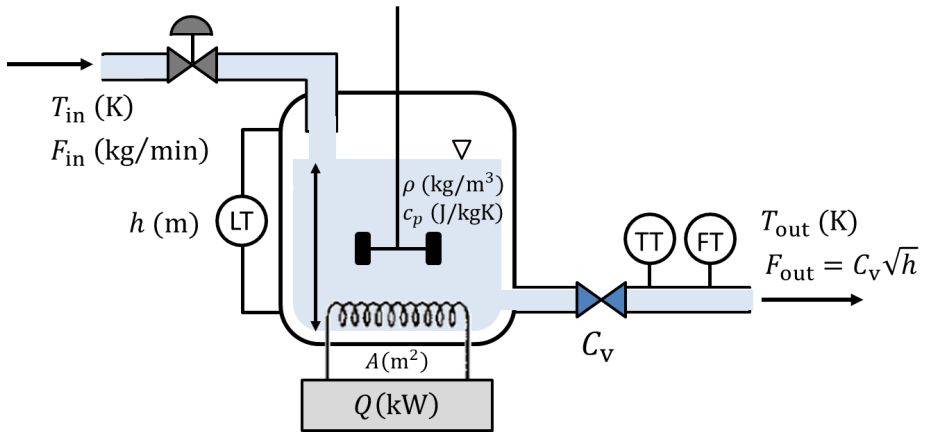
```
% Operating Point
y_nom = [200, 305.17, 2];    % [Fout, Tout, h]

%  State-space Model
A = [-0.05 0; 0 -0.1];
B = [0.001 0 0; -0.003585 0.1 0.0072];
C = [50 0; 0 1; 1 0];
D = zeros(3);
sys = ss(A,B,C,D);

OPT = stepDataOptions('StepAmplitude',...
    [10, 0, 0]);          % [Fin, Tin, Q] Step Input
[Y,T] = step(sys,OPT);    % Simulate step response
Y = Y + y_nom;            % Add the nominal values to outputs

subplot(311); plot(T,Y(:,1,1),'b');      % Plot response of Fout
grid on; ylabel('Fout (kg/min)');
subplot(312); plot(T,Y(:,2,1),'b');      % Plot response of Tout
grid on; ylabel('Tout (K)');
subplot(313); plot(T,Y(:,3,1),'b');      % Plot response of h
grid on; ylabel('h (m)');
```





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MATLAB `step` function

Exercise:

The two tanks in series, as shown, has the following linearized SS model:

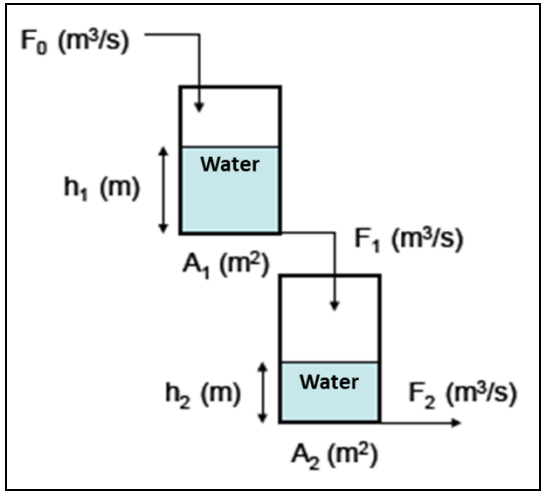
$$\begin{bmatrix} \frac{dh_1'}{dt} \\ \frac{dh_2'}{dt} \end{bmatrix} = \begin{bmatrix} \frac{-\beta_1}{A_1}\left(0.5\bar{h}_1^{-0.5}\right) & 0 \\ \frac{\beta_1}{A_2}\left(0.5\bar{h}_1^{-0.5}\right) & \frac{-\beta_2}{A_2}\left(0.5\bar{h}_2^{-0.5}\right) \end{bmatrix} \begin{bmatrix} h_1' \\ h_2' \end{bmatrix} + \begin{bmatrix} \frac{1}{A_1} \\ 0 \end{bmatrix} F_0'$$

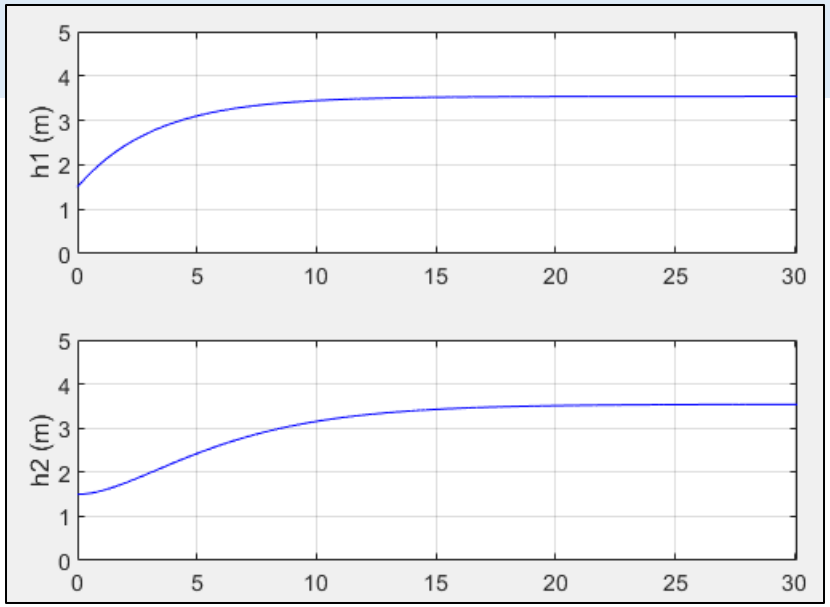
where the outputs are h_1' and h_2', which are the liquid heights in each tank. If the initial operating point is at $\bar{h}_1 = \bar{h}_2 = 1.5$ m, plot the response of both liquid heights to a step change of $+5.0\ m^3/s$ in F_0.

Parameters:

$A_1 = A_2 = 8\ \text{m}^2$

$\beta_1 = \beta_2 = 6\ \text{m}^{2.5}/\text{s}$






```
% Operating point and Constants
y_nom = [1.5 1.5]; % [h1, h2]
beta = 6; area = 8;

% SS model
A = [-beta/area*0.5/sqrt(1.5) 0; ...
     beta/area*0.5/sqrt(1.5) -beta/area*0.5/sqrt(1.5)];
B = [1/area; 0]; C = eye(2); D = zeros(2,1);
sys = ss(A,B,C,D);

OPT = stepDataOptions('StepAmplitude',5);    % F0 Step Input
[Y,T] = step(sys,OPT);                       % Simulate step response
Y = Y + y_nom;                               % Add the nom values to y

subplot(211); plot(T,Y(:,1,1),'b');      % Plot response of h1
grid on; ylabel('h1 (m)'); axis([0 max(T) 0 5]);
subplot(212); plot(T,Y(:,2,1),'b');      % Plot response of h2
grid on; ylabel('h2 (m)'); axis([0 max(T) 0 5]);
```

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Relationship bet. State-space Models and Transfer Functions

State-space Model

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

...can be converted to a **matrix of transfer functions, $\mathbf{G}(s)$.**

$$\mathbf{G}(s) = \mathbf{C}[s\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} + \mathbf{D}$$

Resolvent Matrix

In MATLAB, you can convert between S.S. and T.F. using `ss2tf` and `tf2ss`.

Example:

A simple 2nd order state-space model:

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \underset{\mathbf{A}}{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}} \underset{\mathbf{x}}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} + \underset{\mathbf{B}}{\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}} \underset{\mathbf{u}}{\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \underset{\mathbf{C}}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} \underset{\mathbf{x}}{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} + \underset{\mathbf{D}}{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}} \underset{\mathbf{u}}{\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}}$$

Is exactly equivalent to 4 transfer functions:

$$\frac{\mathbf{Y_1'(s)}}{\mathbf{U_1'(s)}} = \frac{a_{12}b_{21} + b_{11}(s - a_{22})}{(s - a_{22})(s - a_{11}) - a_{21}a_{12}}$$

$$\frac{\mathbf{Y_2'(s)}}{\mathbf{U_1'(s)}} = \frac{a_{21}b_{11} + b_{21}(s - a_{11})}{(s - a_{22})(s - a_{11}) - a_{21}a_{12}}$$

$$\frac{\mathbf{Y_1'(s)}}{\mathbf{U_2'(s)}} = \frac{a_{12}b_{22} + b_{12}(s - a_{22})}{(s - a_{22})(s - a_{11}) - a_{21}a_{12}}$$

$$\frac{\mathbf{Y_2'(s)}}{\mathbf{U_2'(s)}} = \frac{a_{21}b_{12} + b_{22}(s - a_{11})}{(s - a_{22})(s - a_{11}) - a_{21}a_{12}}$$

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Outline

How to formulate MIMO models?

- General Form of a DAE Model
- Model Linearization
- State-space (SS) Models

How to simulate MIMO models?

- Time-Domain Solution of SS Models
- MIMO Systems in MATLAB Simulink

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Solutions of SS Models

The solution to the following continuous linear time-invariant (LTI) SS model…

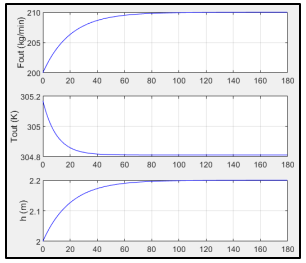
$$\dot{\boldsymbol{x}} = \boldsymbol{Ax} + \boldsymbol{Bu}$$
$$\boldsymbol{y} = \boldsymbol{Cx} + \boldsymbol{Du}$$

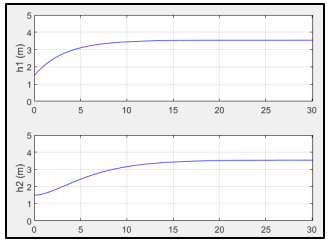
is given by:

Free Response Forced Response

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{Bu}(\tau)\ d\tau$$

$$\boldsymbol{y}(t) = \boldsymbol{Cx}(t) + \boldsymbol{Du}(t)$$





Derivation:

- Consider the single-variable state equation: $\dot{x} = ax + bu$

$$\dot{x} - ax = bu$$

- Multiply an integrating factor, e^{-at}: $e^{-at}\frac{dx}{dt} - e^{-at}ax = e^{-at}bu$
- Notice the following product rule: $\frac{d}{dt}(e^{-at}x) = e^{-at}\frac{dx}{dt} - e^{-at}ax$
- We can now re-write the equation as: $\frac{d}{dt}(e^{-at}x) = e^{-at}bu$
- Perform the integration: $\int_0^t d(e^{-at}x) = \int_0^t e^{-a\tau}bu(\tau)\ d\tau$
- The integral on the left can be evaluated: $e^{-at}x(t) - x(0) = \int_0^t e^{-a\tau}bu(\tau)\ d\tau$
- Re-arranging the equation:

$$x(t) = e^{at}x(0) + e^{at}\int_0^t e^{-a\tau}bu(\tau)\ d\tau$$

$$x(t) = e^{at}x(0) + \int_0^t e^{a(t-\tau)}bu(\tau)\ d\tau$$

- Writing the equation for multivariate, $\boldsymbol{x}$:

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{Bu}(\tau)\ d\tau$$

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Solutions of SS Models

$$\boldsymbol{x}(t) = e^{\boldsymbol{At}}\boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{Bu}(\tau)\, d\tau$$

$$\boldsymbol{y}(t) = \boldsymbol{Cx}(t) + \boldsymbol{Du}(t)$$

Definition:

The **state-transition matrix**, $\boldsymbol{\Phi}(t) = e^{\boldsymbol{At}}$, is defined by the *matrix exponential*:

$$e^{\boldsymbol{At}} = \boldsymbol{I} + \boldsymbol{At} + \frac{\boldsymbol{A}^2t^2}{2!} + \frac{\boldsymbol{A}^3t^3}{3!} + \cdots + \frac{\boldsymbol{A}^kt^k}{k!} + \cdots$$

Given the eigen-decomposition of $\boldsymbol{A}$ as follows:

$$\boldsymbol{A} = \boldsymbol{V\Lambda V}^{-1}$$

the matrix exponential can be computed as:

$$e^{\boldsymbol{At}} = \boldsymbol{V}\exp(\boldsymbol{\Lambda}t)\,\boldsymbol{V}^{-1} = \boldsymbol{V}\begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & e^{\lambda_n t} \end{bmatrix}\boldsymbol{V}^{-1}$$

Important Properties of the Matrix Exponential:

$$\boldsymbol{\Phi}(0) = e^{\boldsymbol{A}0} = \boldsymbol{I}$$

$$\boldsymbol{\Phi}(-t) = e^{-\boldsymbol{At}} = \left[e^{\boldsymbol{At}}\right]^{-1}$$

$$\boldsymbol{\Phi}(t_1 + t_2) = e^{\boldsymbol{A}(t_1+t_2)} = e^{\boldsymbol{A}t_1}e^{\boldsymbol{A}t_2}$$

$e^{(\boldsymbol{A}_1+\boldsymbol{A}_2)t} = e^{\boldsymbol{A}_1 t}e^{\boldsymbol{A}_2 t}$ only if $\boldsymbol{A_1A_2} = \boldsymbol{A_2A_1}$

$$\frac{d}{dt}\boldsymbol{\Phi}(t) = \frac{d}{dt}e^{\boldsymbol{At}} = \boldsymbol{A}e^{\boldsymbol{At}} = e^{\boldsymbol{At}}\boldsymbol{A}$$

$$\int_0^t \boldsymbol{\Phi}(t)\, dt = \int_0^t e^{\boldsymbol{At}}\, dt = \boldsymbol{A}^{-1}\left[e^{\boldsymbol{At}} - \boldsymbol{I}\right] = \left[e^{\boldsymbol{At}} - \boldsymbol{I}\right]\boldsymbol{A}^{-1}$$

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Solutions of SS Models

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{u}(\tau)\, d\tau$$

$$\boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{x}(t) + \boldsymbol{D}\boldsymbol{u}(t)$$

Definition:

The **state-transition matrix**, $\boldsymbol{\Phi}(t) = e^{\boldsymbol{A}t}$, is defined by the *matrix exponential*:

$$e^{\boldsymbol{A}t} = \boldsymbol{I} + \boldsymbol{A}t + \frac{\boldsymbol{A}^2t^2}{2!} + \frac{\boldsymbol{A}^3t^3}{3!} + \cdots + \frac{\boldsymbol{A}^kt^k}{k!} + \cdots$$

Given the eigen-decomposition of $\boldsymbol{A}$ as follows:

$$\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}$$

the matrix exponential can be computed as:

$$e^{\boldsymbol{A}t} = \boldsymbol{V}\exp(\boldsymbol{\Lambda}t)\,\boldsymbol{V}^{-1} = \boldsymbol{V}\begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & e^{\lambda_n t} \end{bmatrix}\boldsymbol{V}^{-1}$$

Example:

Find the analytical solution to the following homogeneous LTI with initial conditions, $x_1(0) = 2$ and $x_2(0) = 3$:

$$\dot{x}_1 = -2x_1$$
$$\dot{x}_2 = x_1 - x_2$$

Solution:

Writing as a state-space model:

$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-2 & 0 \\ 1 & -1\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix}$$

$$\boldsymbol{A} = \begin{bmatrix}-2 & 0 \\ 1 & -1\end{bmatrix}$$

Compute the eigen-decomposition of $\boldsymbol{A}$:

$$\boldsymbol{A} = \begin{bmatrix}-2 & 0 \\ 1 & -1\end{bmatrix} = \begin{bmatrix}-1 & 0 \\ 1 & 1\end{bmatrix}\begin{bmatrix}-2 & 0 \\ 0 & -1\end{bmatrix}\begin{bmatrix}-1 & 0 \\ 1 & 1\end{bmatrix}^{-1}$$

$$\boldsymbol{V} \times \boldsymbol{\Lambda} \times \boldsymbol{V^{-1}}$$

Compute the matrix exponential $e^{\boldsymbol{A}t}$:

$$e^{\boldsymbol{A}t} = \begin{bmatrix}-1 & 0 \\ 1 & 1\end{bmatrix}\begin{bmatrix}e^{-2t} & 0 \\ 0 & e^{-t}\end{bmatrix}\begin{bmatrix}-1 & 0 \\ 1 & 1\end{bmatrix}^{-1}$$

$$e^{\boldsymbol{A}t} = \begin{bmatrix}-e^{-2t} & 0 \\ e^{-2t} & e^{-t}\end{bmatrix}\begin{bmatrix}-1 & 0 \\ 1 & 1\end{bmatrix}^{-1}$$

$$e^{\boldsymbol{A}t} = \begin{bmatrix}e^{-2t} & 0 \\ -e^{-2t} + e^{-t} & e^{-t}\end{bmatrix}$$

Using the formula: $\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0)$, we have,

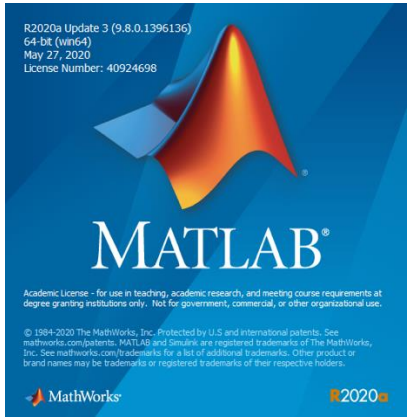
$$\boldsymbol{x}(t) = \begin{bmatrix}e^{-2t} & 0 \\ -e^{-2t} + e^{-t} & e^{-t}\end{bmatrix}\begin{bmatrix}2 \\ 3\end{bmatrix}$$

$$\boldsymbol{x}(t) = \begin{bmatrix}2e^{-2t} \\ -2e^{-2t} + 5e^{-t}\end{bmatrix}$$

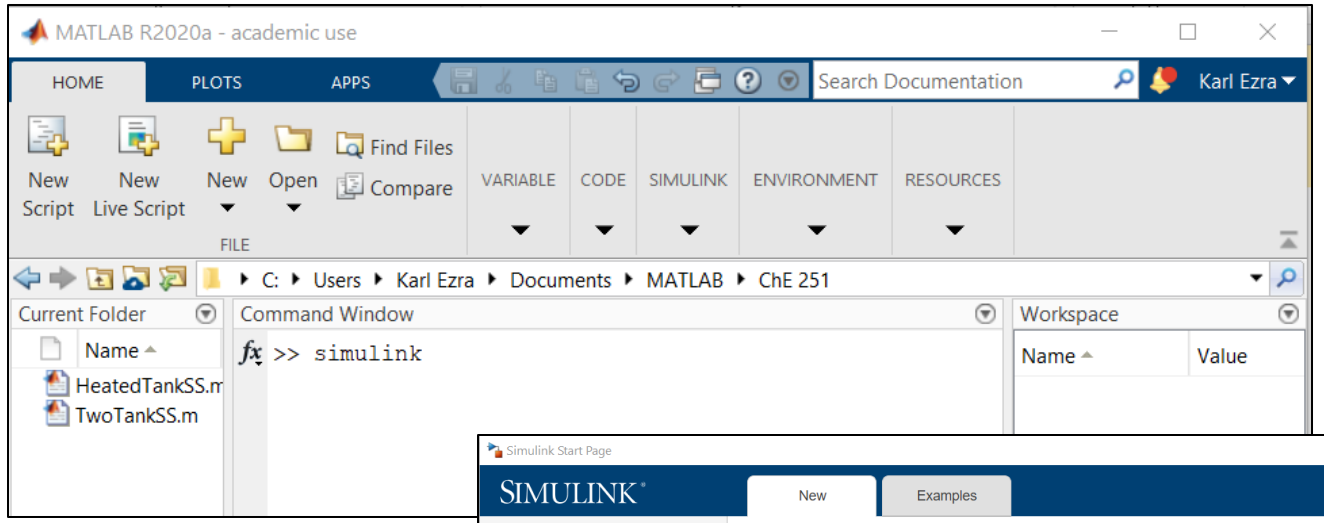
$$x_1(t) = 2e^{-2t}$$
$$x_2(t) = -2e^{-2t} + 5e^{-t}$$

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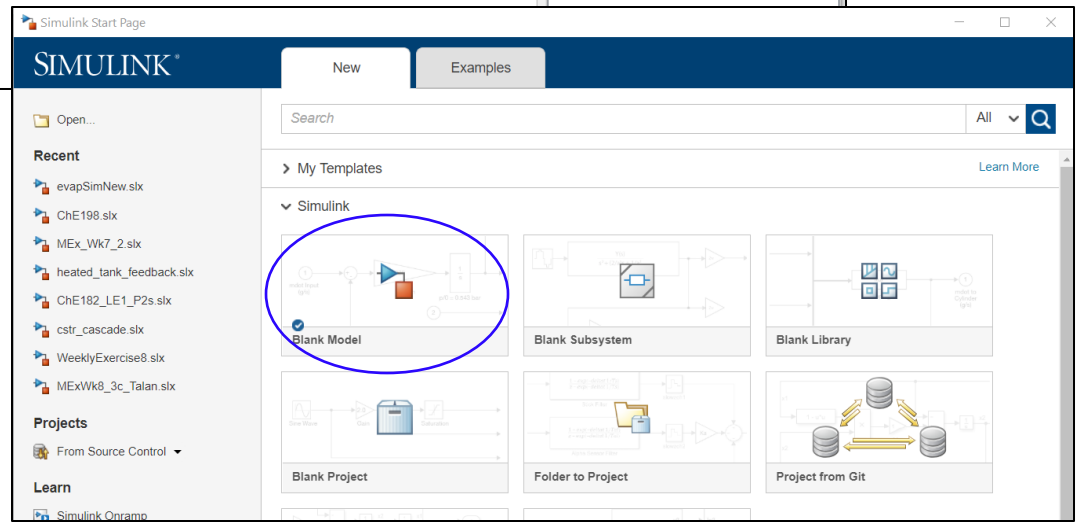
MATLAB Simulink



- Open MATLAB. Type Simulink then press Enter.



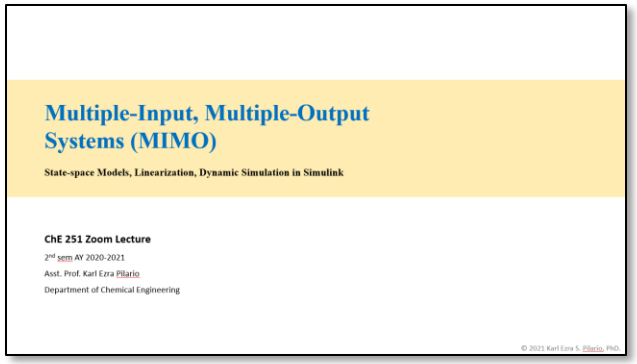
- Click on **Blank Model**.



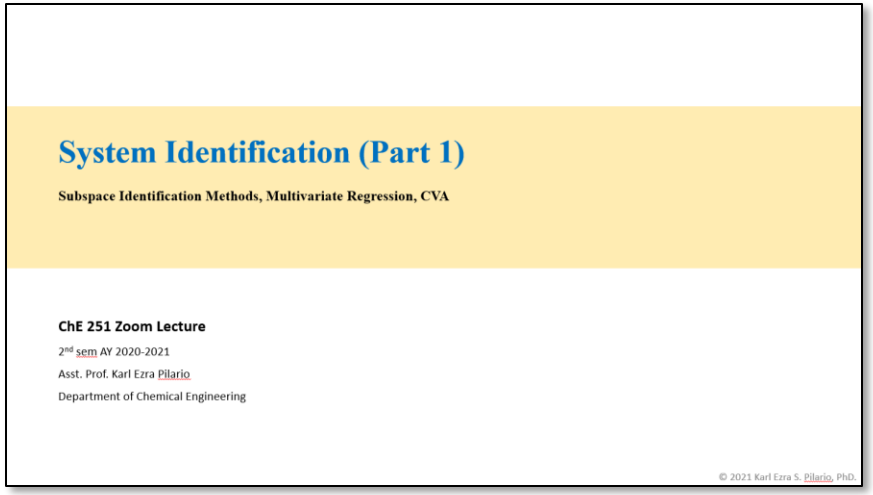
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Outline

- General Form of a DAE Model
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Next week:



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